

SYLLABUS

AI 402 / AI 502: Generative AI

Summer 2026 (May/June Term)

Fully Asynchronous Online

Cross-listed: AI 402 §01 (undergraduate) • AI 502 §01 (graduate)

Credits: 3

Term: May 11, 2026 – June 25, 2026 (6 weeks)

LMS: Blackboard Ultra

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Office hours: 9:00 AM – 3:00 PM ET, Monday–Friday, subject to availability.

Book via Calendly: <https://calendly.com/zachdebruine> (Zoom or in-person — select your preference when booking).

Description

Every industry is figuring out what generative AI is actually good for — and what it isn't. This course puts you in the middle of that question: you will build, ship, and critically evaluate three real projects so you leave with hands-on fluency, not just familiarity.

We focus on what is durable in 2026: prompting, retrieval, agents, tools (including MCP), memory, model routing/waterfalls, and evaluation. We use the best free tools available (ChatGPT, Claude, Gemini, GitHub Copilot, Lovable) and students are welcome to use any additional tool that advances the aims of their work, goals, and projects.

No coding background required. Every project involves using AI to work with code, but coding knowledge or skills are not necessary. Projects offer no-code paths (Lovable, GPT/Claude artifacts) and code paths (Copilot, Claude Code). Pick the path that stretches you. Success in this course is not determined by whether you have paid subscriptions, but students with paid tiers will be able to do more and learn more.

Learning Objectives

After completing this course, students will be able to:

- Explain in plain language how modern LLMs generate text, where their abilities come from, and where they predictably fail.
- Engineer prompts and context (system prompts, few-shot, structured output, retrieval) to make a generative model reliably useful for a defined task.
- Build a retrieval-augmented system that grounds a model in a corpus of your choice.
- Design and ship an agentic workflow that uses tools (function calling and/or MCP), persistent memory, and a model-routing strategy (“model waterfall”).
- Evaluate a generative system with both quantitative metrics and structured qualitative review, and use that evaluation to drive iteration.

- Apply generative AI to a substantive problem in your own discipline and articulate the ethical, legal, and practical limits of your solution.
- Deploy a working application to a public or instructor-accessible hosted environment.

Who This Course Is For

This is a widely interdisciplinary course, with students from domains including AI, data science, cybersecurity, applied computer science, health sciences, communication, business, and more. You are the domain expert in your field; this course gives you the AI fluency. You are encouraged to apply generative AI to problems in your own discipline — the projects you build will be strongest when they reflect your expertise.

Academic Stewardship

All students are expected to be exemplary stewards of the discipline through collegiality, diligence, and integrity, and are expected to encourage others towards similar success. As part of their endeavors, students are to adhere to the academic honesty standards set forth by Grand Valley State University (<https://www.gvsu.edu/computing/academic-honesty-30.htm>), and to the GVSU policies listed at www.gvsu.edu/coursepolicies.

You are required to use generative AI in this course. It is the subject. Specific to this course:

- **Yes:** Use AI to learn, draft, code, debug, brainstorm, critique, summarize.
- **No:** Pasting a project brief into a chatbot and submitting whatever it returns.
- Students will strive to participate actively and creatively, contribute to discussions collaboratively, and think critically about shifting paradigms in this rapidly developing domain.

You do not need to formally disclose AI use on every submission, but reflecting on how you used AI — what worked, what didn't, what you learned — is excellent content for your weekly post-project reflections.

GVSU policies on academic integrity apply.

Required Materials & Accounts

All free unless noted. Set up in Week 0.

Tool	Purpose	Cost
ChatGPT account	General LLM use	Free tier sufficient
Claude.ai account	General LLM use, Projects, Artifacts	Free tier sufficient
Google AI Studio (Gemini)	Long-context experiments, free API key	Free
GitHub + Copilot	Code hosting + code path (free via GitHub Education)	Free for students
Lovable	Full-stack vibe-coding	Free tier

Tool	Purpose	Cost
Blackboard Ultra	Course delivery	—

Optional / instructor-demonstrated: Claude Code (CLI; ~\$5–20 in API credits), OpenAI API (\$5 minimum), Cursor or Windsurf, Google Colab, NotebookLM. Students are welcome to use any tool that advances their work.

No textbook. Readings are short articles, papers, and docs linked from each week’s module. Estimated reading load: 30–60 min/week.

Course Structure

Each week follows a **Monday-to-Sunday** rhythm:

- **Monday:** new module unlocks (lecture videos, notes, project brief, optional readings)
- **Wednesday:** mid-week check-in post + answers to FAQs
- **Sunday 11:59 PM ET:** weekly deliverables due

Three Two-Week Projects (80% of Grade)

The course is built around three intensive, two-week projects that you define. Each project follows a **draft** → **final** cycle.

You choose your own project topics. Work out project questions or ideas with your instructor during office hours or via email (debruinz@gvsu.edu). Office hours can be Zoom or in-person — select your preference when booking.

- **Week 1 (draft):** Graded formatively per rubric item:
 - **Pass (P)** — meets all specifications.
 - **Revise (R)** — close, but needs targeted improvement; feedback identifies exactly what to fix.
 - **Not Yet (NY)** — substantial gaps; significant work needed.
- **Week 2 (final):** Graded summatively per rubric item:
 - **Pass (P)** — meets the specification.
 - **Fail (F)** — does not meet the specification.

No revisions on finals. Bonus points may be awarded at instructor discretion for exceptional work.

Projects are not assigned fixed themes. You design three distinct projects, each applied to a real problem in your own discipline, with a real audience and a real success criterion. The three projects must *collectively* demonstrate the objectives below, but each project must be different from the others in (a) the AI engineering methods it foregrounds, (b) the artifact it produces, and (c) the audience it serves. Repeating the same architecture three times will not pass. Each weekly module releases templates and examples drawn from a range of disciplines; you choose a template or propose your own.

Collective Objectives Across the Three Projects

Across your three projects, you must demonstrate:

- **Prompt engineering** with structured output, few-shot, role + constraints, and step-by-step patterns.
- **System prompt design** that defines persona, scope, and refusal behavior.
- **Retrieval-augmented generation** with a curated corpus, chunking choices, and inline citations.
- **Multi-model use**: at least two providers compared on the same task, and at least one model-routing or model-waterfall design.
- **Tool use** via function calling and/or MCP.
- **Agentic workflows** with planning, multi-step execution, and (in at least one project) sub-agents.
- **Memory** — conversation, knowledge, and persistent — applied where it fits the task.
- **AI-assisted development**: directing GitHub Copilot, Claude Code, or Lovable to build and ship working software; a build log of the prompts used.
- **Evaluation**: a documented eval set, quantitative and qualitative results, and an honest failure analysis.
- **Deployment** to a public hosted environment the instructor can demo.
- **Disciplinary application**: substantive engagement with a problem in your own field, beyond a generic chatbot.
- **Ethical, legal, and practical limits** articulated for what you built.

All projects must be **deployed and fully operational**. Code on GitHub. Deployed to GitHub Pages, Vercel, Lovable, or similar. The instructor must be able to demo a working version.

Discussions (10% of Grade)

All discussions are public and close every **Sunday at 11:59 PM ET**. Six weekly graded discussions.

- **Introduction (Week 0, ungraded)**: Post your name and five things in your area of expertise where AI can create value and solve currently unsolved problems.
- **Weekly method journal**: Post a new method, technique, or workflow you explored that helped you set up an agent, copilot, or achieve a better response. Must be original to the board (or offer an original angle), written to help other students, include a lesson learned, and be at least 100 words.
- **Weekly Q&A**: Post blockers and questions. The instructor reviews and addresses these every Monday.

Reflections (10% of Grade)

200-word maximum text entry on Blackboard each week (6 total). Submitted *after* your project submission. Address: what worked, what didn't, blockers, successes, technologies learned.

Graded Components

Component	Weight
Projects (3 projects × final submission)	80%
Discussions (6 weekly method journals + Q&A)	10%
Reflections (6 weekly)	10%

Graduate Tier (AI 502)

AI 402 and AI 502 students complete the same deliverables. Graduate students are assessed for **deeper critical understanding** on each rubric item—stronger analytical reasoning, more thorough evaluation, and more substantive engagement with underlying concepts and trade-offs.

Response & Availability

- **Email (preferred) / Blackboard messages:** answered as soon as possible. Email: debruinz@gvsu.edu.
- **Project feedback:** returned early in the new week after submission.
- **Discussion Q&A:** reviewed every **Monday**.
- **Office hours:** 9:00 AM – 3:00 PM ET, Monday–Friday, subject to availability. Book via Calendly: <https://calendly.com/zachdebruine>. Zoom or in-person — select your preference when booking.
- I do not respond to messages on Saturdays or Sundays—plan accordingly.

Late Work

Due to the short timeline of this course and the need to focus intensively on each project, **late project submissions will not be accommodated** outside of extreme circumstances. If you are facing an emergency, email the instructor as soon as possible.

Accessibility, Accommodations, Title IX, and University Policies

Students with disabilities who need accommodations should contact Disability Support Resources at <https://www.gvsu.edu/dsr/>. Title IX inquiries: <https://www.gvsu.edu/titleix/>. Mental health support: University Counseling Center, <https://www.gvsu.edu/counsel/>. Religious observance accommodations are available with advance notice. Full university policies: www.gvsu.edu/coursepolicies.

Schedule

- Modules unlock Mondays; projects due Sunday 11:59 PM ET (except Project 3 final: Wed Jun 25).
- Wednesdays are mid-week check-ins and FAQ posts.

Dates	Theme	Due	Project
May 12–18	Foundations & Prompting	P1 Draft Reflection 1 Discussion	Project 1
May 19–25	Retrieval, Memory & Grounding	P1 Final Reflection 2 Discussion	Project 1
May 26–Jun 1	Vibe Coding & AI-Assisted Dev	P2 Draft Reflection 3 Discussion	Project 2
Jun 2–8	Agentic AI I: Tools, MCP, Waterfalls	P2 Final Reflection 4 Discussion	Project 2
Jun 9–15	Agentic AI II: Memory, Orchestration, Eval	P3 Draft Reflection 5 Discussion	Project 3
Jun 16–25	Deployment, Ethics & What's Next	P3 Final (Jun 25) Reflection 6 Discussion	Project 3